

PRAKTIKUM SISTEM OPERASI

MODUL V
EKSPLORASI PROSES



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
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JURNAL

1. Soal 1

Jawablah pertanyaan berikut ini:

- a. Berapa banyaknya maksimum proses yang ada pada Xinu?
Untuk maksimum proses pada Xinu sebanyak: 8 proses, karena diambil dari konstanta NPROC: 8 pada file 'process.h', karena array pada NPROC hanya dapat menampung 8 slot.
- b. Berapa maksimal panjang nama suatu proses pada Xinu?
Untuk panjang maksimal dari proses xinu sendiri adalah 16 karakter, yang dimana field pada 'prname' yang merupakan array char dengan ukuran 16 byte, sehingga proses hanya bisa 16 karakter maksimalnya.
- c. Berapa nilai prioritas awal pada saat proses dibuat?
Nilai prioritas awal pada suatu proses yaitu 20, mengapa 20 dikarenakan pada sebuah nilai yang bernama INITPRIO yang terletak di 'process.h'.
- d. Ada berapa total state pada Xinu? Sebutkan!
Untuk state terdapat 5 state proses yaitu:
 - Free (PR_FREE): Slot proctab kosong / proses tidak aktif (belum dibuat atau sudah diterminasi)
 - Current (PR_CURR): Proses sedang berjalan di CPU (hanya satu proses pada satu waktu)
 - Ready (PR_READY): Proses siap dijalankan, menunggu giliran CPU di ready queue
 - Receive (PR_RECV): Proses sedang menunggu pesan masuk (blocked waiting for message)
 - Sleep / Wait (PR_SLEEP/PR_WAIT): Proses sedang tidur (sleep) atau menunggu semaphore / event tertentu

2. Soal 2

Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

- a. Mencari File 'xsh_ps.c' dengan menggunakan: 'find . -name "xsh_ps.c"' kemudian menggunakan cat

```
Activities Terminal Mon 6 Apr, 04:20
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cat shell/xsh_ps.c
/* xsh_ps.c - xsh_ps */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_ps - shell command to print the process table
 *-----
 */
shellcmd xsh_ps(int nargs, char *args[])
{
    struct procent *prptr;          /* Pointer to process */
    int32 i;                        /* Index into proctabl */
    char *pstate[] = {              /* Names for process states */
        "free ", "curr ", "ready", "recv ", "sleep", "susp ",
        "wait ", "rtime"};

    /* For argument '--help', emit help about the 'ps' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays information about running processes\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
    }
}
```

b. Memberikan komentar dengan menambahkan nama dan juga nim

```
Activities Terminal Mon 6 Apr, 04:39
xinu@xinu-develop-end: ~/xinu/shell

File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh_ps.c Modified

    }

    /* Print header for items from the process table */

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
        "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
        "Stack Ptr", "Stack Size");

    printf("%3s %-16s %5s %4s %4s %10s %-10s %10s\n",
        "___", "-----", "----", "----",
        "-----", "-----", "-----");

    /* Output information for each process */
    /* Nama: Zaenarif Putra Ainurdin, Nim: 2311104049 */
    for (i = 0; i < NPROC; i++) {
        prptr = &proctab[i];
        if (prptr->pstate == PR_FREE) { /* Skip unused slots */
            continue;
        }
        printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
            i, prptr->prname, pstate[(int)prptr->pstate],
            prptr->prprio, prptr->prppid, prptr->prstackbase, prptr->prstackptr, prptr->prstacksize);
    }

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

```

Activities Terminal Mon 6 Apr, 04:41
xinu@xinu-develop-end: ~/xinu

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %10s %-10s %10s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size");

printf("%3s %-16s %5s %4s %10s %-10s %10s\n",
       "----", "-----", "-----", "----", "-----",
       "-----", "-----");

/* Output information for each process */

/* Nama: Zaenarif Putra Ainurdin, Nim: 2311104049 */
for (i = 0; i < NPROC; i++) {

    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->prname, prptr->prstate, prptr->prprio, prptr->prparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen);
}
return 0;
}
xinu@xinu-develop-end:~/xinu$

```

Berikut adalah hasil ketika sudah berhasil memberikan komentar pada file 'xsh_ps.c'

3. Soal 3

Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4
1	rdsproc	susp	200	0	0x005FBFFC	0x005FBFC8	16384	0	0
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0

Kolom Msg adalah banyaknya pesan yang ada dalam proses. Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya

```
Activities Terminal Mon 6 Apr, 04:52
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh ps.c Modified

}

/* Print header for items from the process table */

printf("%3s %-16s %5s %4s %10s %-10s %10s %3s %7s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %10s %-10s %10s %3s %7s\n",
       "----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "----", "-----");

/* Output information for each process */
/* Nama: Zaenarif Putra Ainurdin, Nim: 2311104049 */
for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
           i, prptr->prname, pstate[(int)prptr->prstate],
           ^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
           ^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```

```
Activities Terminal Mon 6 Apr, 04:57
xinu@xinu-develop-end: ~/xinu/shell
File Edit View Search Terminal Help
GNU nano 2.2.6 File: xsh ps.c Modified

/* Output information for each process */
/* Nama: Zaenarif Putra Ainurdin, Nim: 2311104049 */
for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %7d\n",
           i, prptr->prname, pstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen, prptr->prhasmsg
           prptr->prmsg);
    }
    return 0;
}

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```

```

Activities Terminal Mon 6 Apr, 05:00
xinu@xinu-develop-end: ~/xinu/shell

return 1;
}

/* Print header for items from the process table */
printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %7s\n",
       "Pid", "Name", "State", "Prio", "Ppid", "Stack Base",
       "Stack Ptr", "Stack Size", "Msg", "Content");

printf("%3s %-16s %5s %4s %4s %10s %-10s %10s %3s %7s\n",
       "---", "-----", "-----", "-----", "-----", "-----",
       "-----", "-----", "-----", "-----");

/* Output information for each process */

/* Nama: Zaenarif Putra Ainurdin, Nim: 2311104049 */
for (i = 0; i < NPROC; i++) {
    prptr = &proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d %3d %7d\n",
           i, prptr->prname, prptr->prstate[(int)prptr->prstate],
           prptr->prprio, prptr->prparent, prptr->prstkbase,
           prptr->prstkptr, prptr->prstklen, prptr->prhasmsg,
           prptr->prmsg);
}

```

- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.
- Screenshot source kode dan output akhir hasil modifikasi

```

Activities Terminal Mon 6 Apr, 05:08
xinu@xinu-develop-end: ~/xinu/compile

-----2311104049-----
XINU
-----ZAENARIF PUTRA 'AINURDIN-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE90  8192  1    3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30  8192  0    0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0  8192  0    0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536  0    0
5 shell           recv   50   4 0x005D7FFC 0x005D79AC  8192  0    0
6 ps              curr   20   5 0x005F7FFC 0x005F7DD8  8192  0    0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

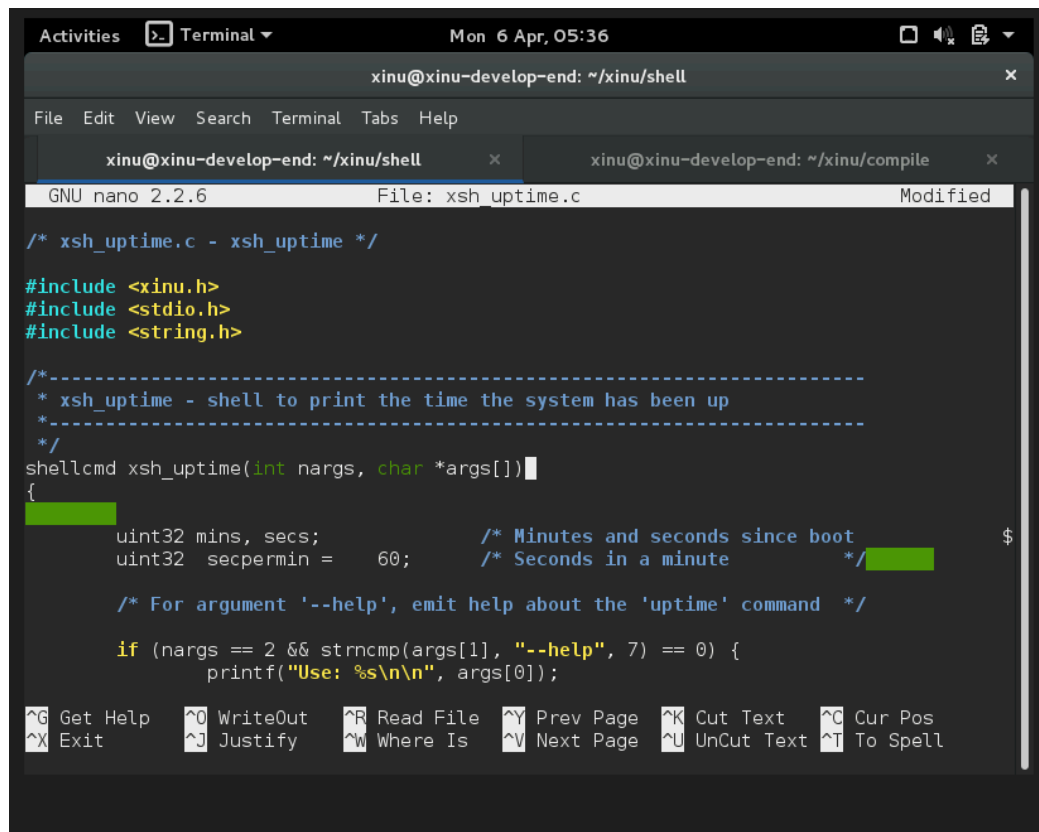
```

4. Soal 4

Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source kode dan output akhir hasil modifikasi



The screenshot shows a terminal window titled "Terminal" with the date and time "Mon 6 Apr, 05:36". The terminal is running the nano text editor, editing the file "xsh_uptime.c". The code visible in the editor is as follows:

```
/* xsh_uptime.c - xsh_uptime */
#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32 mins, secs;          /* Minutes and seconds since boot
    uint32 secpermin = 60;      /* Seconds in a minute */

    /* For argument '--help', emit help about the 'uptime' command */

    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
    }
}
```

The terminal window also shows a menu of keyboard shortcuts at the bottom, such as ^G Get Help, ^O WriteOut, ^R Read File, ^Y Prev Page, ^K Cut Text, ^C Cur Pos, ^X Exit, ^J Justify, ^W Where Is, ^V Next Page, ^U UnCut Text, and ^T To Spell.

Activities Terminal Mon 6 Apr, 05:37

xinu@xinu-develop-end: ~/xinu/shell

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/shell xinu@xinu-develop-end: ~/xinu/compile

GNU nano 2.2.6 File: xsh_uptime.c Modified

```
printf("Options:\n");
printf("\t--help\t display this help and exit\n");
return 0;
}

/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

secs = clktime;          /* Total seconds since boot */

/* Calculate total minutes */
mins = secs / secpermin;

/* Print the uptime strictly in minutes*/
```

Get Help WriteOut Read File Prev Page Cut Text Cur Pos
Exit Justify Where Is Next Page UnCut Text To Spell

Activities Terminal Mon 6 Apr, 05:41

xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

```
xsh $ uptime
Xinu has been up 0 minute(s)
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0